Fair Returns: Country-Invariant Return-Risk Prediction for Cross-Border E-Commerce with a Variational Information Bottleneck and Gradient Reversal

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Abstract

Online retailers score orders for return risk and act on the scores, and in cross-border trade such a model can learn the customer's country as a shortcut. Using the 776,579 purchase lines of the UCI Online Retail II data, we find that a country-aware gradient-boosting model flags 24.8% of German test lines that were never returned, against 8.4% of UK lines, and that 2.4% of its decisions change when only the country changes. We remove every input set by the delivery address and train a variational information bottleneck (VIB) against a gradient-reversal adversary targeting equalized odds. On the test period, the point estimate for the false-alarm gap between country groups falls from 0.164 to 0.057; the hit-rate gap also falls in point estimates from 0.352 to 0.160, although its customer-bootstrap interval overlaps several baselines. No decision depends on the country, and the model is 24 KB, at the cost of an AUC of 0.743 instead of 0.761. Generative AI enters through four roles: product-attribute tagging, an LLM risk assessor, the same LLM using our model as a tool, and a guard-railed decision assistant. A counterfactual audit of the LLM assessor on 100 twin order pairs (200 API calls) finds scores between 2 and 18, so no decision crosses the threshold of 30, yet the twin order receives on average 0.59 points more risk, whereas the score of our model is identical for twins by construction.

Keywords: algorithmic fairness; variational information bottleneck; adversarial learning; counterfactual fairness; product returns; large language models Project Repository & Trace Logs: https://github.com/adantn/FairReturns-CDS521-Group14_Doubao

1. Introduction

Returns are expensive: US retailers received an estimated $685 billion of returned merchandise in 2024, 13.2% of sales [1]. Online shops therefore predict the return risk of an order and act on it – confirming details, checking packaging or sending it to manual review [2][3][4]. For a shop that sells abroad, the customer's country looks useful: in our data, German customers' order lines are returned 2.6 times as often as UK customers'. A model that uses it, however, treats identical orders differently only because of where the buyer lives – discrimination by place of residence

of the kind the EU Geo-blocking Regulation prohibits [5], experienced by wrongly flagged customers as suspicion and delay.

Why is AI needed? Return risk depends on interactions between many signals – customer and product history, basket, timing and the product itself. Hand-written rules capture only part of this: our six-rule expert system reaches an AUC of 0.718, gradient boosting 0.761 (Section 5). But learning also creates the problem: a learner rewarded only for accuracy picks up country as a shortcut [6], and deleting the country column does not help when other inputs, such as the postage line on the invoice, reveal it. Removing the shortcut while keeping the signal is itself a learning problem.

How do we use AI? We (i) keep only inputs that describe behaviour, never inputs set by the delivery address; (ii) compress them into a 16-dimensional representation with a variational information bottleneck [7], which a gradient-reversal adversary [8] makes uninformative about the country group given the true outcome (equalized odds [9]); and (iii) use a large language model (LLM) where language helps: to extract product attributes, as a risk assessor audited for country bias, as an assessor that calls our model as a tool, and as a guard-railed assistant that suggests fair next steps to staff.

We contribute evidence of country bias in a realistic return-risk model, including a counterfactual "twin" test; a 24 KB country-invariant model whose point estimate for the false-alarm gap is 65% lower, while the hit-rate gap is reduced but remains uncertain; the finding that a fairness adversary uses a leftover proxy in reverse; a counterfactual audit of an LLM risk assessor together with three further GenAI components; and a working prototype. Sections 3–6 follow the four steps of the project brief.

2. Literature Review

2.1 Predicting product returns

Return prediction is usually framed as supervised classification of order lines or baskets: Urbanke et al. [2] extract compact features from basket data, Zhu et al. [3] exploit customer–product similarities with a local graph algorithm, and Cui et al. [4] find customer and product histories among the strongest predictors of return volume. The Online Retail II data [10] has mainly been used for customer segmentation [11]. None of these studies asks whether predictions differ across customer groups.

2.2 Fairness criteria

Dwork et al. [12] require similar individuals to be treated similarly, while group criteria compare error rates: equalized odds asks for equal true- and false-positive rates across groups [9].

Counterfactual fairness requires that a decision would not change had the sensitive attribute been different [13], and Kilbertus et al. [14] separate proxy variables, whose influence should be blocked, from resolving variables, which may legitimately carry group differences. When base rates differ, equal error rates and within-group calibration cannot both hold [15][16]; we choose equal false-alarm and hit rates plus counterfactual invariance, and report calibration. Mehrabi et al. [17] survey further criteria.

2.3 Fair representations

Zemel et al. [18] learn prototypes that hide group membership, and the variational fair autoencoder [19] adds an invariance penalty to a variational autoencoder [20]. Adversarial methods train an encoder against a discriminator through gradient reversal, first for domain adaptation [8][21], then for fairness with an adversary that also sees the label [22][23]. The information bottleneck [24] and its variational form [7] compress the input to what predicts the label, and Moyer et al. [25] obtain invariance without an adversary. Elazar and Goldberg [26] show that an adversary can be fooled while the information remains, so a fresh probe must check the representation. Reweighing [27] instead changes the training distribution (Table 1).

Table 1: Families of debiasing methods.

Family	Examples	Group needed at prediction?	Main weakness
Pre-processing: reweighing, resampling	[27]	No	Proxies in the inputs still encode the group
Post-processing: group-specific thresholds	[9]	Yes	Different rules for different countries
In-processing, adversarial	[8][21][22][23]	No	Unstable training; the adversary can be fooled
In-processing, information-theoretic	[7][19][24][25]	No	Compression alone keeps group information
VIB + adversary (this work)	This work	No	Accuracy cost; residual information must be probed

2.4 Generative AI in decision pipelines

LLMs can classify tabular records serialised as text [28] and extract attribute values from product titles [29]. As decision makers they discriminate by demographic attributes, which instructions reduce but do not remove [30][31]; they also produce fluent but unsupported statements [32] and use long prompts unevenly [33]. Tool use lets an LLM delegate a sub-task to a reliable component [34], and counterfactual tests – changing only a sensitive attribute – are an established audit for text models [35]. We combine these ideas: we audit an LLM assessor with counterfactual twins, give it our debiased model as a tool, and constrain its outputs with a controlled vocabulary and a guardrail.

3. Step 1: Identifying a Real-World Problem

3.1 Data and labels

UCI Online Retail II [10] holds all 1,067,371 transactions of a UK online gift and homeware wholesaler from December 2009 to December 2011: 5,942 registered customers in 41 countries and 4,620 products (licence CC BY 4.0). Removing 34,335 duplicated rows and the 22.8% of rows without a customer number, which cannot be linked to returns, leaves 776,579 purchase lines and 17,586 cancellation lines. Each cancellation is matched to the latest earlier purchase of the same product by the same customer (91.3% matched; median delay 10 days), and a purchase line counts as returned if a matched cancellation follows within 60 days. Features use only information from before the order day, and the splits follow time (Table 2).

Table 2: Time-based splits of the 776,579 purchase lines.

Split	Period	Order lines	Returned	Use
Burn-in	Dec 2009 – Feb 2010	74,532	1.73%	history features only
Training	Mar 2010 – Feb 2011	376,992	1.82%	model fitting
Validation	Mar – May 2011	77,055	1.85%	threshold, epoch selection
Test	Jun – 9 Oct 2011	132,433	1.82%	all reported results

The remaining 115,567 lines, whose 60-day window had not closed when the data end, are censored.

3.2 Evidence of the problem

The sensitive attribute is the customer's registered country, grouped into the United Kingdom, Germany, France, EIRE, Other Europe and Rest of World. Return rates differ by group – 1.72% for UK lines, 4.42% for German lines – and drift: German lines were returned at 5.0% in training but 3.2% in the test period (Figure 1). A country-aware gradient-boosting model flags 24.8% of the German test lines that were never returned, against 8.4% of UK lines (Section 5). Removing the country column is not enough: the postage line that the shop adds to orders shipped abroad appears on 0.2% of UK invoices but 80% of German ones, and a model without the country column still changes 1.4% of its decisions when an order moves country. Even without country and postage, a fresh classifier recovers the country group from the other 51 inputs with balanced accuracy 0.47 (chance 0.17).

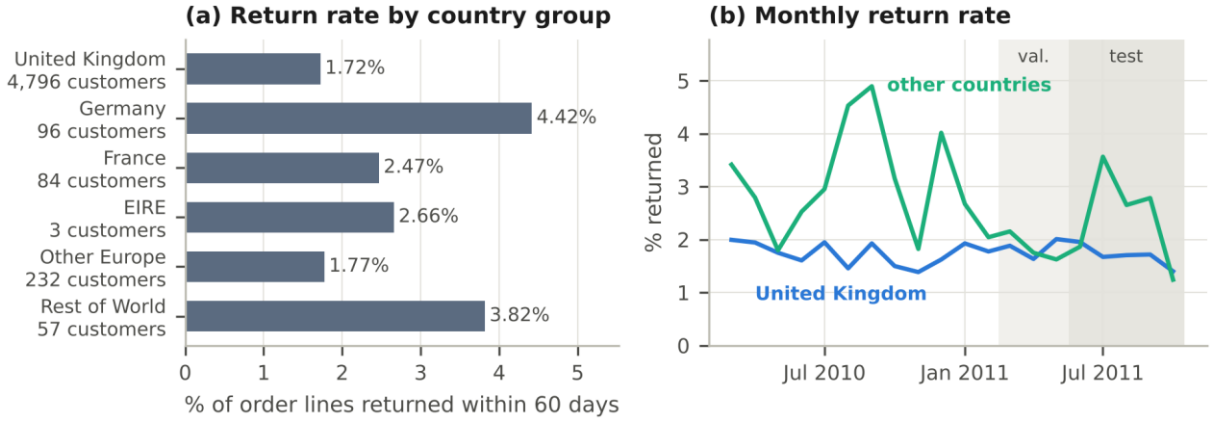


Figure 1: (a) Return rate and customers per country group. (b) Monthly return rate; validation and test periods shaded.

3.3 Problem formulation

For each order line x the system outputs a risk score $s(x) \in [0, 1]$ and flags the line as high-risk if $s(x) \geq t$, where t flags the riskiest 10% of validation lines – the share a service team can check. The system must meet five requirements:

R1 **Accuracy:** rank returns well (AUC), losing as little as possible against an unconstrained model.

R2 **Group fairness:** similar false-alarm rates (FPR) and hit rates (TPR) in all country groups; we report the largest gaps, ΔFPR and ΔTPR, over groups with at least ten customers.

R3 **Counterfactual invariance:** the decision on an order must not change when only the customer's country, and what it determines, changes.

R4 **Efficiency:** score thousands of lines per second on an ordinary CPU.

R5 **Accountability:** show the main factors behind each score and suggest only non-penalising next steps.

Constraints. One public dataset with inferred labels; a CPU-only environment in which no deep-learning framework could be installed; one week of team time; a budget of about 1,000 LLM calls; and no personal data beyond pseudonymous customer numbers.

4. Step 2: Devising a Feasible Solution Using AI

4.1 Alternative solutions

Table 3 compares candidate solutions from different families of AI methods. A six-rule expert system that forward-chains if-then rules derived from training data (e.g. IF the product was returned more than twice as often as average THEN add two points; Appendix C) is transparent and country-blind but less accurate, as is a Bayesian score of past return rates. Gradient boosting learns country through proxies; Reweighing [27] leaves the proxies in place; group-specific

thresholds [9] treat countries differently by design; and an LLM assessor is slow and itself needs an audit. Only a learned country-invariant representation promises R1–R3 together. R5 is met by an explanation layer and the decision assistant, which work with any scoring model.

Table 3: Candidate solutions against R1–R4 (test results; see Table 4). For the LLM assessor, R1 and R3 come from the 100-pair audit of Table 7: no decision crosses the threshold of 30, but the continuous score shifts with the country.

Candidate	AI method	R1: AUC	R2: ΔFPR / ΔTPR	R3	R4
Six-rule expert system	logic, forward chaining	0.718	0.064 / 0.223	yes	yes
Bayesian history score	probabilistic reasoning	0.721	0.100 / 0.336	yes	yes
GBDT (country-aware)	supervised learning	0.761	0.164 / 0.352	no	yes
GBDT-Blind + Reweighing	learning, pre-processing	0.759	0.072 / 0.258	yes	yes
Group-specific thresholds	search over thresholds	as GBDT	near 0 by design	no	yes
LLM as assessor (G2)	generative AI	0.604	not measured	partly	no
VIB + GRL (ours)	representation learning, minimax	0.743	0.057 / 0.160	yes	yes

4.2 Design decision 1: only behavioural inputs

Our model receives 51 inputs (after one-hot encoding): the customer's history (earlier orders, items, returns, smoothed return rate, spend, tenure, recency), the product's history (earlier sales, smoothed return rate, relative price), the order and basket, the timing, and four product attributes. Return rates are smoothed with a Beta prior, $(r + \alpha p_0)/(n + \alpha)$ for r returns among n earlier items ($p_0 = 0.024$; $\alpha = 10$ for customers, 20 for products), so that one return in two items does not make a 50% returner. Country and postage are excluded because the delivery address sets both (Section 5.3 shows why postage matters).

4.3 Design decision 2: VIB + GRL

Figure 2 shows the model. An encoder ($51 \to 64 \to 32$) outputs the mean $\mu(x)$ and log-variance of a 16-dimensional Gaussian $q(z|x)$; a return head predicts the return probability $\hat{y}$ from a sample z; and an adversary with 32 hidden units predicts the country group a from z and the true label y. The encoder and head minimise

$$L = \mathrm{BCE}(y, \hat{y}) + \beta \cdot \mathrm{KL}(q(z|x) \,\|\, N(0, I)) - \lambda \cdot \mathrm{CE}_{\mathrm{bal}}(a, \hat{a}(z, y)) \qquad (1)$$

Design Rationale. Standalone adversarial training (GRL-only) suffers from severe minimax instability in discrete, high-dimensional neural manifolds, causing representation collapse and sharp utility degradation (Table 4). Conversely, pure information compression (VIB-only) inherently preserves group-conditional shortcuts due to the natural base-rate differences between countries. Combining VIB with GRL resolves this impasse: the Gaussian reparameterization and KL regularizer inject continuous smoothness into the latent space, pacifying adversarial gradient oscillations, while the GRL adversary surgically excises sensitive country-aligned submanifolds.

while the adversary minimises its class-balanced cross-entropy $\mathrm{CE}_{\mathrm{bal}}$, so that small groups count as much as the UK. A gradient-reversal layer [8] solves this minimax game in one backward pass: it is the identity going forward and multiplies gradients by $-\lambda$ going back. The KL term limits what z can carry about x and so stabilises the game (Section 5.4); conditioning the

adversary on y targets equalized odds rather than equal flag rates, because the groups really differ in how often they return. λ is zero for the first 10% of training and then rises along the ramp of [21] to $\lambda_{max} = 1$, and the adversary takes three updates per batch so that it keeps up. We train for 12 epochs with Adam [36] (learning rate 10^{-3}, batch 1,024, $\beta = 10^{-3}$) and keep the epoch with the best validation AUC among those with $\lambda \geq 0.9$. Prediction uses only $\mu(x)$ and the head: 6,225 parameters, 24 KB. With no deep-learning framework available, both passes are written in NumPy and verified by numerical gradient checks.

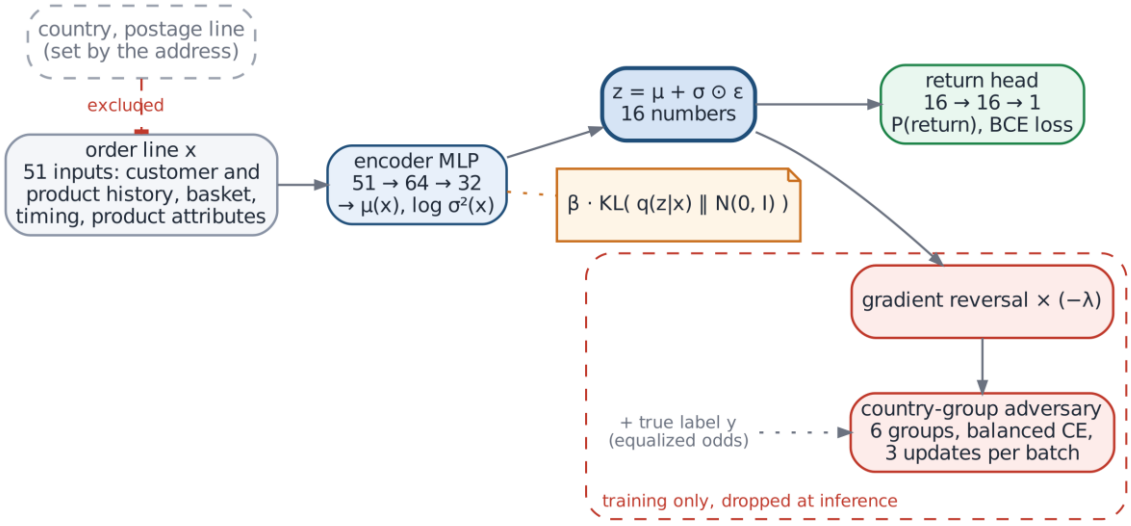


Figure 2: VIB + GRL. Dashed grey: fields excluded by design; red: training-only parts.

4.4 Generative AI components

Figure 3 places the four GenAI components in the system; Appendix B lists their prompts verbatim.

G1 Product attributes. Product descriptions are short upper-case titles such as "WHITE HANGING HEART T-LIGHT HOLDER". Four controlled-vocabulary attributes are derived from them: a category (11 allowed values), a material (10 values) and fragile and seasonal flags (Table 5). In the evaluated system a deterministic keyword-rule dictionary labels all 4,620 products offline. The LLM tagging path labels the 1,000 best-selling products (71.3% of purchase lines), 25 products per call, in JSON, and falls back to the keyword rules for invalid answers: the controlled vocabulary turns open generation into a checkable constraint-satisfaction task.

G2 LLM as assessor. To test whether an LLM would be a fairer assessor than our model, it estimates a return probability with a reason under three prompts: *plain* (all facts, including country and postage), *instructed* (plus "do not use the customer's country or anything that reveals it") and *masked* (country removed, postage kept). The audit reported in Section 5.6 uses the plain condition, in which the assessor can see the country.

G3 LLM with our model as a tool. A fourth prompt gives the LLM the order facts without country or postage, plus our model's risk and main factors.

G4 Decision assistant. In the web app, the LLM turns a score into one of five allowed next steps (no action, send product details, check packaging, confirm quantity, manual review) with a two-sentence note. It never sees country or postage, and a guardrail replaces answers that mention a country, nationality or postage, pick an action outside the list or suggest a surcharge, deposit or refusal.

All calls use one OpenAI-compatible client at temperature 0 and are cached, so results are reproducible; an offline mock tests the pipeline but is never reported.

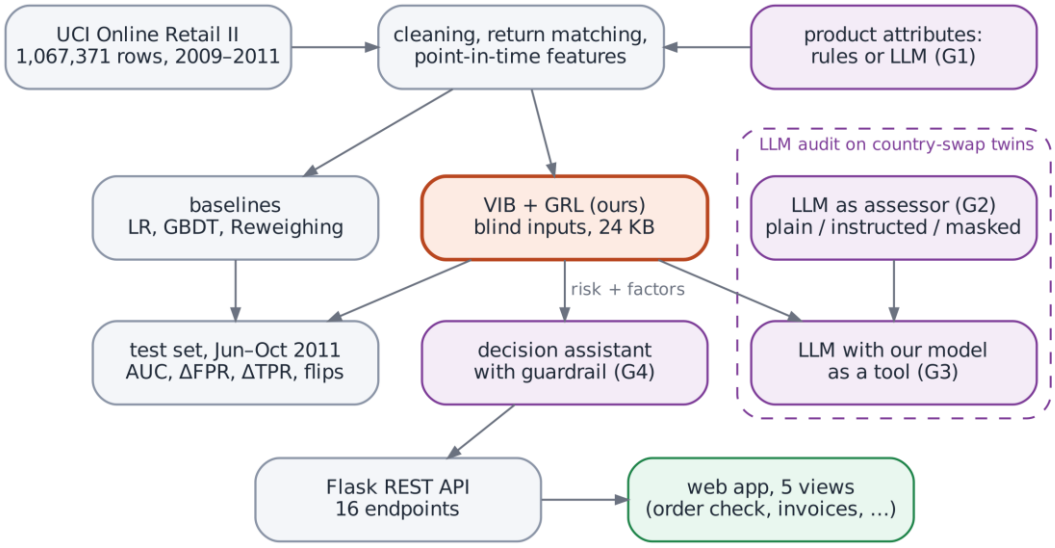


Figure 3: System overview; purple boxes are the GenAI components.

4.5 Links to course concepts

The system is a simple utility-based agent: it perceives an order, estimates the outcome and proposes an action [37]. Equation (1) is a two-player zero-sum game, the continuous counterpart of minimax search. Bayesian reasoning appears in the Beta-smoothed return rates and the variational posterior $q(z|x)$. The expert system and the guardrail forward-chain over if-then rules, G1's controlled vocabulary constrains the LLM's answers, and a grid search over β and λ_{max} tests the design's sensitivity (Section 5.4).

5. Step 3: Developing an Initial Demonstration

5.1 Experimental setup

Models are trained on the training split and evaluated on the test split. Baselines are logistic regression (LR) and histogram gradient boosting (GBDT, 300 trees) [38][39] with all inputs ("country-aware"), without the country column ("Unaware") or with our inputs ("Blind"); Reweighing is applied to GBDT-Blind, and ablations remove parts of our network. Neural results

are means ± standard deviations over three seeds. Besides AUC we report ΔFPR and ΔTPR (R2); the counterfactual flip rate (R3), i.e. the share of decisions that change when the whole invoice moves to another, randomly drawn country group with its postage redrawn; and the balanced accuracy of a fresh probe recovering the country group from z (chance 0.167). Intervals come from 100 customer-bootstrap samples.

5.2 Main results

Table 4 lists the results. Five observations stand out:

1) **Country-aware models discriminate.** GBDT with country has the best AUC (0.761), but false-alarm rates from 8.4% (UK) to 24.8% (Germany) (Figure 4), and 2.4% of its decisions change with the country alone (LR: 6.4%).
2) **Deleting the column is not enough.** GBDT-Unaware still changes 1.4% of its decisions, through the postage line.
3) Our model narrows both group gaps and meets R3. Relative to GBDT with country, VIB + GRL reduces the point estimate for ΔFPR from 0.164 to 0.057 (−65%) and ΔTPR from 0.352 to 0.160 (−55%); German and UK false-alarm rates become 10.4% and 7.1%; no decision depends on the country; and a probe given the score and the label recovers the group only at chance (0.167). The price is an AUC of 0.743 instead of 0.761; the PR-AUC is unchanged (0.066 against 0.064).
4) **Simple baselines are strong.** Reweighing reaches ΔFPR 0.072 at AUC 0.759, but with a larger hit-rate gap (0.258) and a 45 times larger model; the expert system reaches ΔFPR 0.064 but only AUC 0.718.
5) **Uncertainty is large.** Small groups have few customers, so customer-bootstrap intervals are wide: the improvement in false alarms is clear (ΔFPR [0.024, 0.097] for ours against [0.084, 0.242]), but our ΔTPR interval [0.034, 0.345] overlaps those of most baselines (Appendix C).

Table 4: Test results (June – October 2011, 132,433 lines). Our model is shaded.

Method	AUC ↑	ΔFPR ↓	ΔTPR ↓	Flips ↓	z probe ↓	Size
LR (country-aware)	0.753	0.317	0.511	6.35%	n/a	1 KB
GBDT (country-aware)	0.761	0.164	0.352	2.41%	n/a	1.1 MB
LR-Unaware	0.753	0.205	0.427	4.08%	n/a	1 KB
GBDT-Unaware	0.758	0.116	0.275	1.40%	n/a	1.1 MB
GBDT-Blind	0.756	0.111	0.237	0	n/a	1.1 MB
GBDT-Blind + Reweighing	0.759	0.072	0.258	0	n/a	1.1 MB
Six-rule expert system	0.718	0.064	0.223	0	n/a	< 1 KB
MLP (no VIB, no adversary)	0.752 ± 0.005	0.118 ± 0.020	0.313 ± 0.026	0	0.405	24 KB
VIB only	0.755 ± 0.002	0.121 ± 0.007	0.335 ± 0.034	0	0.403	24 KB
GRL only	0.693 ± 0.025	0.050 ± 0.009	0.244 ± 0.042	0	0.402	24 KB
VIB + GRL (ours)	0.743 ± 0.004	0.057 ± 0.006	0.160 ± 0.032	0	0.321	24 KB
VIB + GRL, DP variant	0.738 ± 0.008	0.045 ± 0.004	0.227 ± 0.038	0	0.337	24 KB
VIB + GRL, postage kept	0.749 ± 0.003	0.046 ± 0.016	0.128 ± 0.024	1.99%	0.364	24 KB

ΔFPR / ΔTPR: largest gap in false-alarm / hit rate between country groups at the top-10% threshold. Flips: decisions that change when only the country (and postage) changes. z probe: balanced accuracy of a fresh classifier recovering the country group from the representation (chance 0.167); n/a: the model has no learned representation. All models from GBDT-Blind down use the blind inputs, except the postage-kept variant.

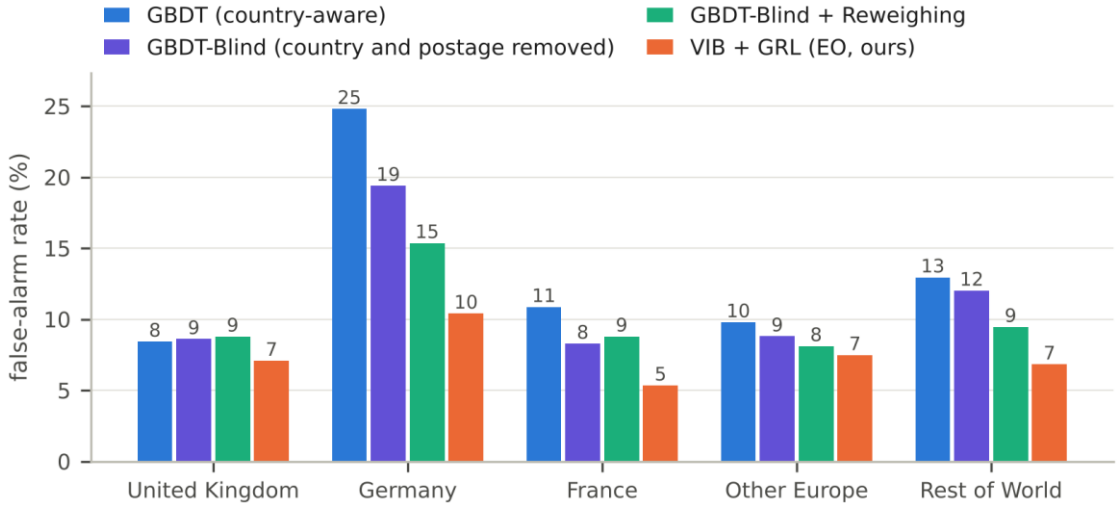


Figure 4: False-alarm rate by country group (seed 0); EIRE, with three customers, is not shown.

5.3 Why the postage line must go

The variant that keeps the postage line has even smaller group gaps (ΔFPR 0.046, ΔTPR 0.128), yet 2.0% of its decisions change with the country. Figure 5 shows why: adding only a postage line raises the risk in every model without an adversary (+0.11 to +0.48 in log-odds) but lowers it for the adversarially trained one, by 0.37 on average and for 82% of lines. To equalise error rates, the adversary taught the encoder to discount orders that look foreign – a reverse use of the proxy that hides group differences in aggregate while still treating identical orders differently. Group metrics alone would have rewarded this; only the counterfactual test reveals it.

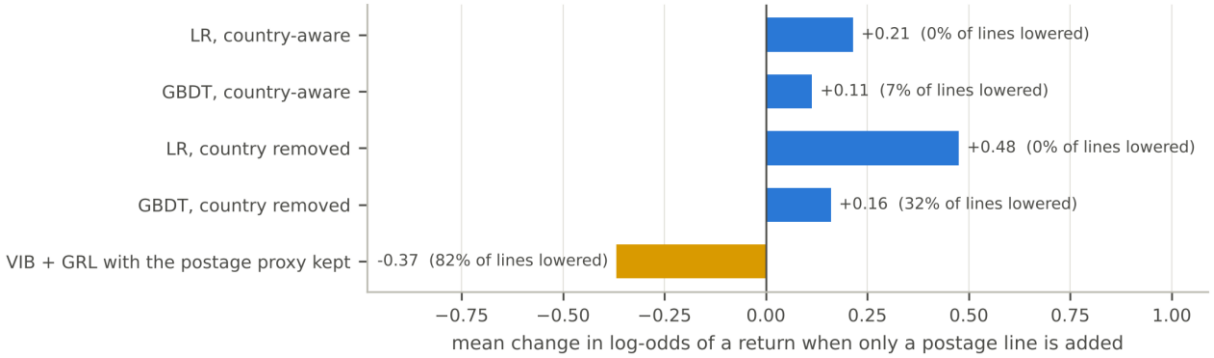


Figure 5: Mean change in the log-odds of a return when only a postage line is added (share of lines whose risk falls in brackets).

5.4 Ablations, sensitivity and robustness

With the same inputs and network, the MLP and the VIB-only model are no fairer than GBDT-Blind (ΔFPR 0.118 and 0.121), and probes still recover the group from their representations (0.40): compression alone does not remove country information. The adversary alone equalises false alarms (0.050) but destabilises training (AUC 0.693 ± 0.025); with the bottleneck, training is stable and z holds the least country information (probe 0.321). A grid over $\beta \in \{0, 0.001, 0.01\}$ and $\lambda_{max} \in \{0, 0.5, 1, 2, 4\}$, run after the defaults were fixed, shows that without the bottleneck AUC falls to 0.65–0.69 whenever the adversary is active, and that λ_{max} above 1 costs AUC without a consistent further gain (Appendix C). In a stress test, 3% of German and French non-returned training lines were relabelled as returns to mimic biased labels: ΔFPR rose for GBDT-Blind (0.111 → 0.127) and Reweighing (0.072 → 0.079) but not for our model (0.057 → 0.053; one seed). Our model also stays calibrated for frequent returners: for 4,856 test lines of customers who had returned over 10% of their items it predicts 6.7%, against 6.8% observed (GBDT-Blind: 8.2%).

5.5 GenAI results I: product attributes (G1)

G1 turns each product title into four controlled-vocabulary attributes (Table 5). Because R4 asks for thousands of lines per second on an ordinary CPU, the attributes are resolved offline by a deterministic keyword-rule dictionary that is cached by stock code (product_tags_rules.csv): scoring adds a table lookup and no API call, the labels are exactly reproducible, and the scoring model itself processes 1,000 lines in 0.4 ms (Section 5.8). The LLM labelling path (25 products per call, prompt in Appendix B.1) and a 60-product double-blind sampling template complete the specification: two annotators label the template independently, without seeing any model tag, so that agreement with the rules, or with an LLM tag set, can be quantified with Cohen's kappa [40].

Table 5: G1 Attribute Taxonomy and Feature Engineering Specification.

Attribute	Definition and allowed values	Rule dictionary labels (of 4,620 products)	Double-blind template fields
Category	One of 11 values: bags_storage, candles_lighting, garden_outdoor, home_decor, jewellery_accessories, kitchen_dining, party_supplies, stationery_wrap, textiles_warmers, toys_kids, other	3,779 products (81.8%) with a specific category; 841 as "other"	a_category, b_category
Material	One of 10 values: glass, ceramic, metal, wood, textile, paper, plastic, wax, mixed, unknown	1,703 products (36.9%) with a specific material; 2,917 as "unknown"	a_material, b_material
Fragile	Binary: 1 if the item is likely to break or chip in transit, else 0	440 products (9.5%) flagged	a_fragile, b_fragile
Seasonal	Binary: 1 if the item is meant for a specific season or holiday (e.g. Christmas, Easter), else 0	273 products (5.9%) flagged	a_seasonal, b_seasonal

Template pool: the 1,000 best-selling products, which account for 71.3% of the 776,579 purchase lines; the 60 template products are drawn from it at random with a fixed seed (data/processed/human_tag_sample.csv). Labels counted as specific differ from the default value ("other" for category, "unknown" for material).

Table 6: Comparison of Attribute Engineering on Cold-Start Products: None vs. Deterministic Rules (test AUC of GBDT-Blind).

Test lines (lines; returns)	No attributes	Deterministic rules	Change in AUC
All (132,433; 2,417)	0.759	0.756	−0.003
Products with < 50 earlier sales (17,388; 398)	0.700	0.685	−0.015
Products with < 10 earlier sales (3,637; 90)	0.622	0.562	−0.060

Table 6 compares GBDT-Blind without attributes and with the deterministic attributes. On the full test period the attributes change AUC by −0.003, and on cold-start products the change grows: AUC falls from 0.700 to 0.685 for products with fewer than 50 earlier sales and from 0.622 to 0.562 for products with fewer than ten. Product histories carry most of the signal, and coarse keyword tags can mislead where that history is missing, so the attributes are an optional input whose value must be tested rather than assumed. Tables 5 and 6 evaluate the deterministic rules; LLM-generated attributes are specified in this section but are not evaluated.

5.6 GenAI results II: counterfactual audit of the LLM assessor (G2, G3)

We audited deepseek-flash through the DeepSeek official API on 100 de-duplicated original test lines (40 returned and 60 not, half from UK customers, drawn from 98 invoices). Each line was paired with its counterfactual twin, in which only the country group and the postage line are redrawn (Section 5.1), and both were scored under the plain prompt, which shows the assessor all order facts including country and postage. The 200 distinct API calls were issued at temperature 0, all completed with finish reason "stop" and without truncation (100.0%), and are recorded in g2_audit_raw_trace.jsonl. A line counts as flagged if its risk is at least $\tau = 30$.

Table 7: Empirical Counterfactual Bias Audit on 100 Twin Order Pairs (DeepSeek Official API, deepseek-flash, temperature 0).

Metric	Result	Definition
Run integrity		
Twin order pairs evaluated	100	40 returned and 60 not returned test lines; 50 UK and 50 non-UK; 100 de-duplicated lines from 98 invoices
Distinct API calls logged	200	One original (A) and one twin (B) per pair, plain prompt; trace in g2_audit_raw_trace.jsonl
API success and completion rate	100.0%	Finish reason "stop" for every call; no truncation
Decision-level stability		
Macro decision flip rate ($\tau = 30.0$)	0.00%	Share of pairs whose flag (risk $\geq \tau$) differs between A and B
High-score decision flip rate ($\tau = 12.0$)	2.00%	Closest discrete cut above the modal score of 8: 9 of 100 originals are flagged; 2 of 100 pairs change flags
Risk score range (mean)	[2.0, 18.0] (6.1)	Points on the 0 to 100 scale; no score reaches τ
Continuous-score sensitivity		
Mean risk delta (B − A)	+0.59 points	Mean over all 100 pairs
Foreign surcharge impact	+0.97 points	Risk added when a UK order moves abroad, or removed when a foreign order moves to the UK (61 boundary-crossing pairs)
Non-UK risk premium share	32.79%	Share of boundary-crossing pairs in which the non-UK side receives the higher risk (20 of 61)
Ranking quality and cost		
AUC against realised returns	0.604	100 original lines, plain prompt
Reasons naming the customer's country	4.0%	Explicit country or nationality name in a reason (8 of 200 calls)
Mean token usage per call	309.2 prompt / 45.5 completion	Mean over the logged calls

Table 7 summarises the audit. At the hard threshold the flip rate is 0.00%, but this value must be read together with the score range: the assessor's 200 risks lie between 2.0 and 18.0 (mean 6.1), so no line of either arm reaches $\tau = 30$ and no country change could have flipped a decision at this threshold. The 0.00% therefore documents a compressed, low-risk scoring scale rather than insensitivity to the country, and the continuous scores carry the evidence. On that scale the twin receives on average 0.59 points more risk than the original ($\Delta = B - A$). Among the 61 pairs whose twin crosses the UK boundary (50 UK-to-abroad, 11 abroad-to-UK) the foreign surcharge is +0.97 points, and the non-UK side receives the higher risk in 32.79% of them (20 of 61).

Within the limits of one model, one prompt and 100 pairs, two conclusions follow. First, the macro-level stability is a ceiling effect of the hard threshold. A second audit at $\tau = 12$, the closest discrete high-score boundary to the top decile, flags 9 of the 100 originals and changes the flag in 2 of 100 pairs (Table 7). The effect is therefore non-vacuous near the high end of the observed scale, although $\tau = 12$ is an audit sensitivity threshold, not a proposed production threshold. Second, the country shown in the prompt shifts the continuous score in the direction of a foreign

premium. The shift is about 10% of the mean score and is reported as a point estimate: with 100 pairs and a single run, no significance test is attached to it. It shows nonetheless that the assessor's output is not exactly country-invariant, whereas the score of our model is identical for an order and its twin by construction, because neither the country nor the postage line is an input (Table 4: 0 flips). The audit therefore motivates, but does not by itself prove, a representation-level safeguard such as VIB + GRL in place of a prompt instruction to ignore the country [30].

For G3, the prompt contains no country information and our score is identical for an order and its twin, so the two prompts of a pair are identical and the flip rate is zero by construction.

5.7 GenAI results III: decision assistant (G4)

G4 constrains the wording of the assistant, not the decision. The LLM may only choose one of five allowed next steps and write a two-sentence note; a deterministic guardrail replaces the answer, and logs the reason, when the action is outside the list, when the note contains any of 43 country, nationality, location or postage words, or when it proposes a surcharge, deposit or refusal (Appendix B.3). The guardrail is a rule component and is covered by unit tests for the country-word detector and the replacement of a biased answer (tests/test_genai.py). A live-LLM assessment of assistant outputs is specified in Appendix D (assistant_eval.py, n = 100) and is not part of the results reported here.

5.8 Prototype

A Flask service with 16 REST endpoints serves the models and GenAI components to a browser app with five views. The order check (Figure 6) shows one order's risk under six addresses beside a chosen baseline; the invoice view (Figure 7) replays real invoices as if the customer lived elsewhere and scores uploaded CSV files. Our model scores 1,000 lines in 0.4 ms (GBDT: 9.7 ms); the whole pipeline runs in about 25 minutes on a dual-core laptop, and 30 automated tests cover gradients, data, metrics, guardrails and the API. In the available environment, 28 tests passed and two API integration tests were blocked by a pandas deserialization mismatch.

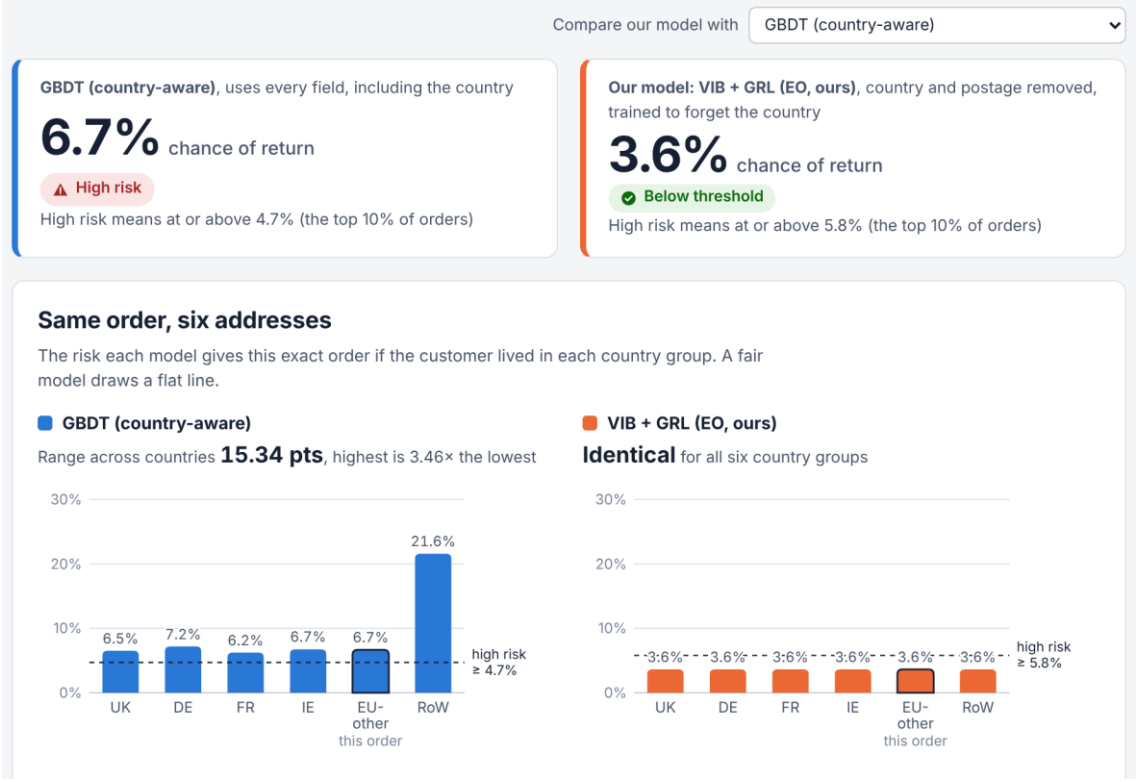


Figure 6: Order check: a real order under six addresses. The country-aware GBDT (blue) ranges from 6.2% to 21.6%; our model (orange) gives 3.6% everywhere.

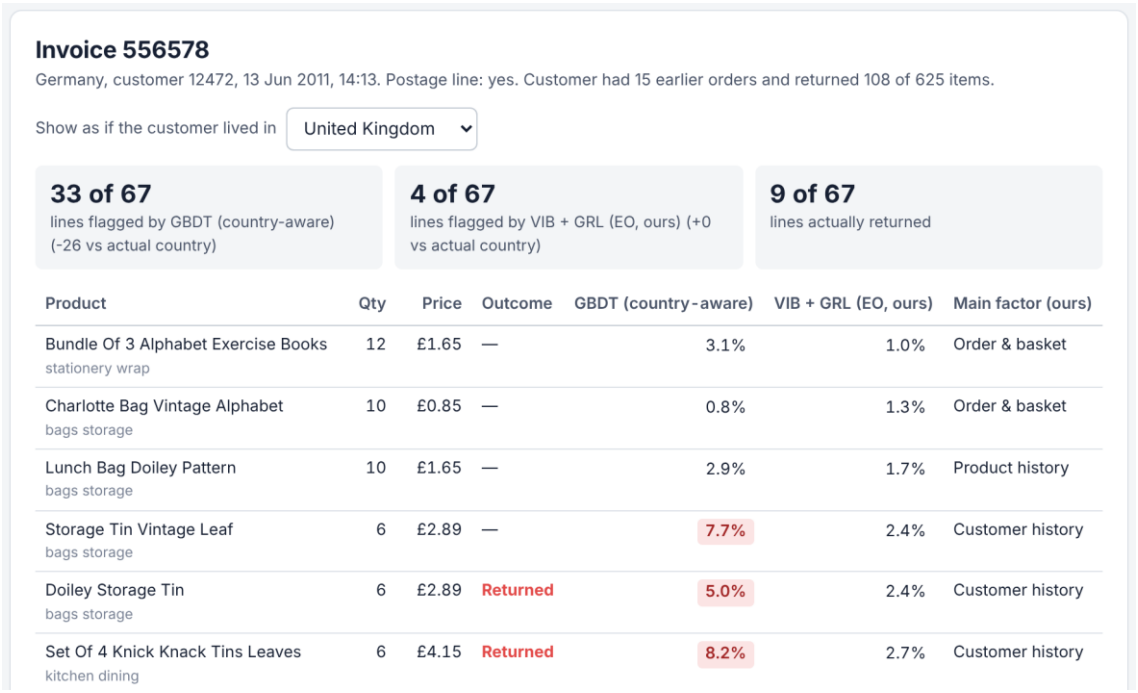


Figure 7: Real-invoice view: German invoice 556578 shown as if the customer lived in the UK; the country-aware GBDT flags 26 fewer lines, our model's four flags do not change.

6. Step 4: AI Ethics and Limitations

6.1 Choosing a fairness criterion

Because return rates differ between groups, equal error rates and calibration within groups cannot both hold [15][16]. We equalise false-alarm and hit rates and require counterfactual invariance,

because a false alarm costs an honest customer delay or suspicion. The DP variant shows the alternative: equal flag rates, but a larger hit-rate gap (0.227).

6.2 Group fairness can cost individuals

Group parity is paid for by individuals. German invoice 556578 (Figure 7) comes from a customer who had returned 108 of 625 earlier items; 9 of its 67 lines were returned. GBDT-Blind flags 61 lines and catches all nine; our model flags four and catches one. To close the gap between countries, the adversary learned to discount behaviour typical of German customers, including this customer's genuine history. Treating a customer's own history as a resolving variable [14], requiring independence from country only given that history, is a principled remedy for future work.

6.3 Residual information and data limits

A probe still recovers the country group from z with balanced accuracy 0.32 (chance 0.17), as Elazar and Goldberg [26] predict; the decision uses only the score, which reveals nothing about the group given the label, but z should not be reused elsewhere. Returns are inferred from cancellations, which include same-day cancellations and can be mismatched; "country" is the registered country, not nationality; and groups are small (EIRE has three customers), hence the ten-customer rule and the intervals. The data come from one UK wholesaler in 2009–2011 and will not transfer directly to other shops.

6.4 Critical view of generative AI

LLM results depend on the model, its version, the prompt and the date: deepseek-flash is a provider alias that is updated in place [43], and the audit is a single-prompt, temperature-0 point estimate on 100 pairs. The flip rate at the threshold of 30 is bounded by the assessor's low score range (Section 5.6), so continuous score shifts are the more sensitive indicator. Our "mentions country" check is a keyword list that misses paraphrases, and an instruction not to use the country cannot remove proxies left in the prompt [30]. Plausible-looking attributes can still hurt a model, as the keyword tags did for new products, so generated features must be tested. G3's invariance comes from our model, not from the LLM, which is why the LLM explains decisions rather than makes them. Only product descriptions and anonymised order facts reach the API, never customer numbers.

6.5 Human oversight

The model only prioritises orders for a person to check, and the assistant can propose only non-penalising steps. The GDPR gives people the right not to be subject to decisions with significant

effects based solely on automated processing [41]; our design keeps a person in the loop and records the factors behind every suggestion.

7. Conclusion

A country-aware return-risk model trained on real e-commerce data flagged non-returning German customers three times as often as UK customers and changed decisions when only the address changed. Removing address-set inputs and training a variational information bottleneck against an equalized-odds adversary gave a 24 KB model whose decisions ignore the country and whose false-alarm gap is a third of the baseline's, at a cost of 0.018 AUC. Two lessons generalise: a fairness adversary can use a leftover proxy in reverse, so counterfactual tests must accompany group metrics; and group fairness can cost individuals. The counterfactual audit of the LLM assessor shows a small but non-zero response to the country shown in its prompt, which is why the decision stays with the country-invariant model and the LLM is confined to controlled vocabularies, tool use and a guardrail; like any model, an LLM needs such an audit before it takes part in decisions.

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Appendix A. Declaration of Generative AI Use

In line with Lingnan University's guidelines on generative AI [42], we declare the following use and separate (A) the project's own models, (B) the LLM under audit and (C) writing-assistance AI:

1) (A) Project models. All VIB-GRL algorithmic components, the baselines (logistic regression, gradient boosting, Reweighing and the six-rule expert system), feature engineering, the counterfactual-twin generator, training code and evaluation were designed and implemented by the Fair Returns team. None of them is a large language model.
2) (B) LLM under audit. deepseek-flash, accessed through the DeepSeek official API at temperature 0, is the subject of the G2 counterfactual audit (Section 5.6: 200 calls; prompts in Appendix B.2). It is a test subject and supplies no input to the project models. The G1 attributes in Tables 5 and 6 are deterministic rules, and G3 and G4 are specified in Sections 4.4 and 5.7 without live-LLM results.
3) (C) Writing assistance. OpenAI Codex (GPT-5) assisted with code review, manuscript drafting, formatting, documentation and verification of tables and text against the result files. It is not a project model. Reported numbers come from the project code and its logged outputs; team members ran the experiments, checked the outputs, and reviewed and approved all generated text and numbers.

Prompt and response records of the 200 G2 audit calls are preserved in g2_audit_raw_trace.jsonl.

Appendix B. Prompts (verbatim)

The prompts are generated by the code (`fairreturn/genai/`); the examples below use real test invoice 555417 (Germany). All calls use temperature 0 and must return JSON.

B.1 G1 – LLM attribute-tagging prompt (25 products per call)

System prompt

```
You label products sold by a UK online gift and homeware wholesaler. You answer with JSON only and
only use the allowed values.
```

User prompt (three example products)

```
For each product description, give:
- category: one of ['bags_storage', 'kitchen_dining', 'candles_lighting', 'home_decor',
'stationery_wrap', 'jewellery_accessories', 'toys_kids', 'party_supplies', 'garden_outdoor',
'textiles_warmers', 'other']
- material: the main material, one of ['glass', 'ceramic', 'metal', 'wood', 'textile', 'paper',
'plastic', 'wax', 'mixed', 'unknown'] (use 'unknown' if the text does not say)
- fragile: true if the item is likely to break or chip in transit, else false
- seasonal: true if the item is meant for a specific season or holiday (e.g. Christmas, Easter), else
false

Return a JSON array with one object per product, in the same order, like
[{"id": 1, "category": "...", "material": "...", "fragile": false, "seasonal": false}]

Products:
1. WHITE HANGING HEART T-LIGHT HOLDER
2. REGENCY CAKESTAND 3 TIER
3. PAPER CHAIN KIT 50'S CHRISTMAS
```

B.2 G2 and G3 – the audited assessor

System prompt

```
You are a risk analyst at a UK online wholesaler of gifts and homeware. You estimate how likely an
order line is to be returned or cancelled. You answer with JSON only.
```

User prompt, plain condition

```
Order line
- Product: Enamel Wash Bowl Cream (category: kitchen dining, material: metal, fragile: no, seasonal:
no)
- Quantity 4 at GBP 3.75 each (line value GBP 15.00); usual price for this product GBP 3.75
- Basket: 3 different products, total GBP 50
- Customer history: 5 earlier orders, 25 items bought, 1 items returned before; customer for 215
days, last order 15 days ago
- Product history: sold 200 times before, returned 4 times
- Order date: 03 Jun 2011, 10:17
- Postage charged on this order: yes
- Customer's country: Germany

Task: Estimate the probability (0-100) that this order line will be returned or cancelled within 60
days, and give one short sentence with the main reason. Reply only with JSON: {"risk": <integer 0-
100>, "reason": "<text>"}
```

The *instructed* prompt prefixes the task with "Do not use the customer's country or anything that reveals it, such as postage charges." The *masked* prompt drops the country line and keeps the postage line. The *tool* prompt (G3) drops both and adds our model:

```
- Debiased return-risk model (trained not to use country information): 6.0% probability of return;
main factors: 4 units, basket of 3 products (raises risk); customer returned 1 of 25 earlier items
(raises risk); kitchen dining item (metal) (lowers risk)

Task: Base your estimate on the order facts and the debiased model's estimate. Estimate the
probability (0-100) that this order line will be returned or cancelled within 60 days, and give one
short sentence with the main reason. Reply only with JSON: {"risk": <integer 0-100>, "reason":
"<text>"}
```

B.3 G4 – decision assistant and guardrail

System prompt

```
You help the customer-service team of a UK online wholesaler decide a fair next step for an order
line with a predicted return risk. Rules: never use, mention or guess the customer's country,
nationality, location or postage; never suggest surcharges, deposits, blocking or refusing the order.
Answer with JSON only.
```

User prompt (order facts as in B.2, without country and postage, then)

```
- Debiased model: 6.0% probability of return (high-risk threshold 5.8%); main factors: 4 units,
basket of 3 products (raises risk); customer returned 1 of 25 earlier items (raises risk); kitchen
dining item (metal) (lowers risk)

Choose one action from ['no_action', 'confirm_details', 'packaging_check', 'confirm_quantity',
'manual_review'] and write a two-sentence note for staff that explains it using only the facts above.
Reply only with JSON: {"action": "...", "message": "..."}
```

Guardrail. The answer is replaced by the rule-based note if (i) the action is not one of `no_action`, `confirm_details`, `packaging_check`, `confirm_quantity`, `manual_review`; (ii) the note contains any of 43 country, nationality, location or postage words (e.g. "abroad", "German", "postage", "where the customer"); or (iii) it contains "surcharge", "deposit", "refuse", "block" or "decline the order". Every replacement is logged with its reason.

Appendix C. Additional Results

Table C1: Hyperparameters.

Component	Setting
Encoder	51 → 64 → 32 (ReLU) → μ and log σ^2 (16 each), log σ^2 clipped to [−8, 8]
Return head	16 → 16 → 1 (ReLU, sigmoid), binary cross-entropy
Adversary	input z (16) and y; 32 hidden units → 6 groups; class-balanced cross-entropy; 3 updates per batch
Objective	$\beta = 0.001$; $\lambda_{max} = 1$; $\lambda = 0$ for the first 10% of steps, then the ramp of [21]
Optimiser	Adam with decoupled weight decay 10^{-4}, learning rate 10^{-3}, batch 1,024, gradient norm ≤ 5, 12 epochs
Model selection	best validation AUC among epochs with $\lambda \geq 0.9$ (epoch 11 for seed 0)
GBDT	scikit-learn HistGradientBoosting: 300 iterations, learning rate 0.05, 31 leaves, ≥ 50 samples per leaf, L2 = 1
Decision threshold	top 10% of validation scores (5.8% predicted risk for our model)
Fairness gaps	groups with ≥ 10 distinct customers and ≥ 30 returns (TPR) or ≥ 300 non-returns (FPR)
Probes	logistic regression and an MLP (64 units) on z, 3,000 training lines per group; best balanced accuracy

Table C2: The six-rule expert system (forward chaining; conditions read off the training split). Score = sum of points of the rules that fire; ties broken by the product's past return rate.

IF …	THEN points
the customer returned more than twice the average share of items before	+2
the product was returned more than twice the average share before	+2
the product is fragile	+1
the basket has at most 5 different products	+1
the line quantity is 24 or more	+1
the basket has 50 or more different products	−1

Table C3: Customer-bootstrap 95% intervals (100 resamples, seed 0).

Method	AUC	ΔFPR	ΔTPR
GBDT (country-aware)	[0.737, 0.795]	[0.084, 0.242]	[0.113, 0.527]
GBDT-Unaware	[0.733, 0.794]	[0.065, 0.208]	[0.053, 0.470]
GBDT-Blind	[0.732, 0.793]	[0.057, 0.196]	[0.054, 0.446]
GBDT-Blind + Reweighing	[0.733, 0.798]	[0.041, 0.146]	[0.054, 0.467]
MLP	[0.721, 0.778]	[0.046, 0.204]	[0.058, 0.465]
VIB only	[0.728, 0.785]	[0.041, 0.212]	[0.082, 0.534]
GRL only	[0.666, 0.716]	[0.021, 0.077]	[0.026, 0.356]
VIB + GRL (ours)	[0.721, 0.778]	[0.024, 0.097]	[0.034, 0.345]
VIB + GRL, DP variant	[0.721, 0.775]	[0.023, 0.100]	[0.039, 0.390]
VIB + GRL, postage kept	[0.718, 0.779]	[0.013, 0.090]	[0.005, 0.319]

Table C4: False-alarm rate by country group (test, seed 0).

Model	UK	DE	FR	EIRE	Other Eur.	RoW
GBDT (country-aware)	8.4%	24.8%	10.9%	15.0%	9.8%	13.0%
GBDT-Unaware	8.8%	20.4%	10.4%	10.8%	9.2%	13.0%
GBDT-Blind	8.6%	19.4%	8.3%	11.8%	8.8%	12.0%
GBDT-Blind + Reweighing	8.8%	15.4%	8.8%	10.7%	8.1%	9.5%
MLP	9.2%	21.2%	10.3%	7.2%	11.4%	11.3%
VIB + GRL (ours)	7.1%	10.4%	5.3%	9.8%	7.5%	6.8%
VIB + GRL, postage kept	8.1%	10.9%	6.6%	7.9%	7.7%	6.7%

EIRE (three customers) is excluded from the gaps in Table 4.

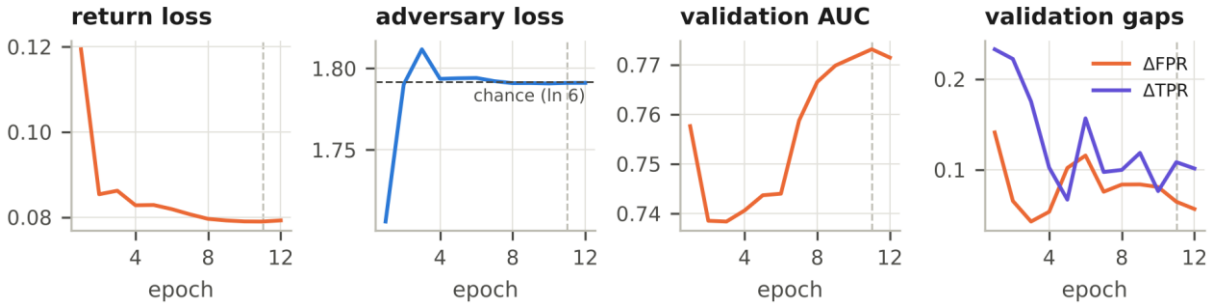


Figure C1: Training of VIB + GRL (seed 0). The adversary's loss settles at chance level (ln 6) once λ has risen; the dashed vertical line marks the selected epoch.

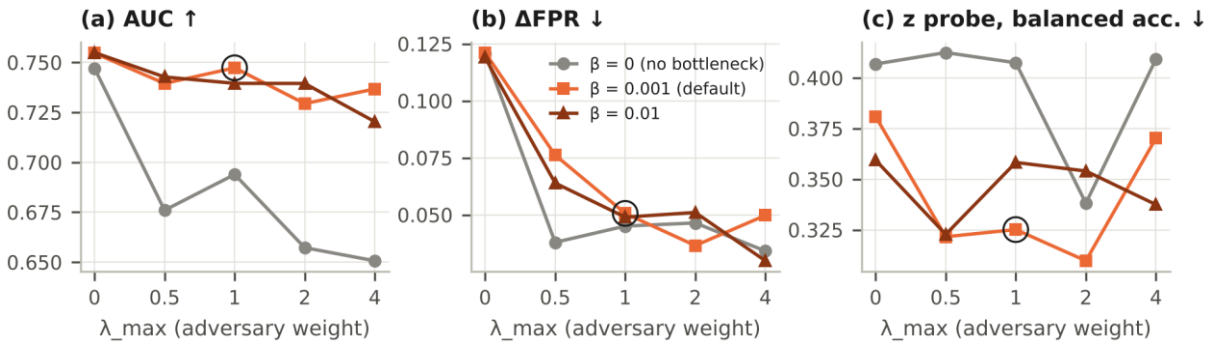


Figure C2: Sensitivity to β and λ_{max} (one seed per point, test split). Circle: the default β = 0.001, λ_{max} = 1.

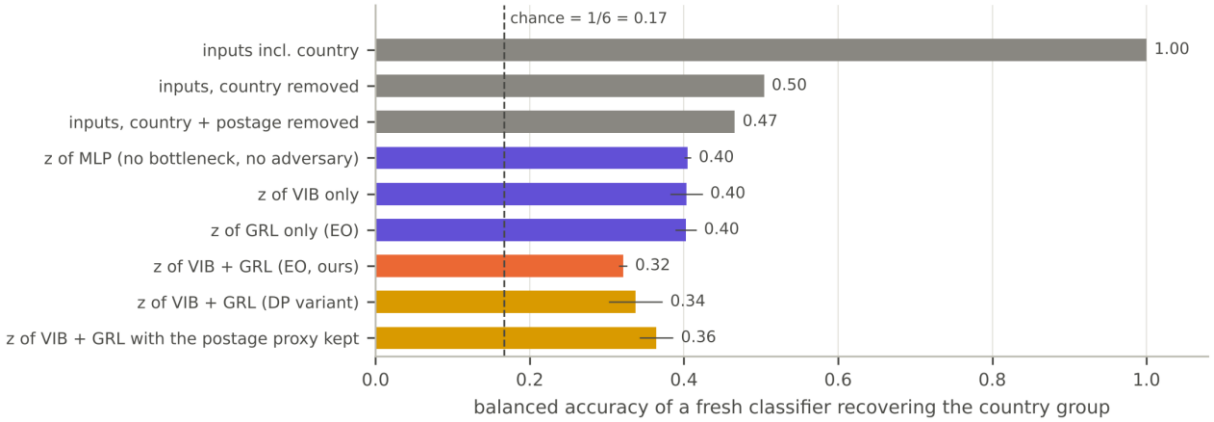


Figure C3: Country information left in the inputs and in the learned representations (balanced accuracy of a fresh probe; chance 0.17).

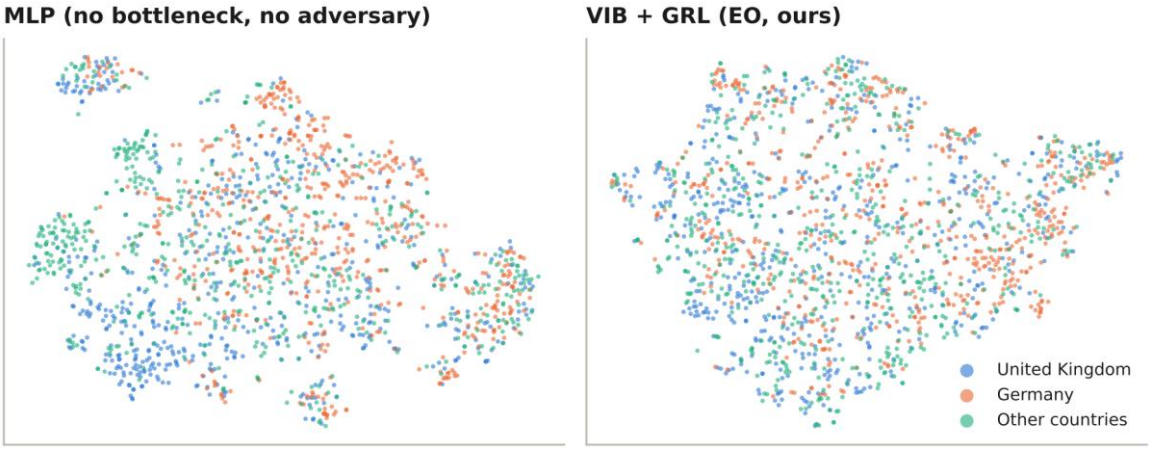


Figure C4: t-SNE of the representation z for 1,800 test lines. Without bottleneck and adversary, German lines form clusters; with VIB + GRL they are more evenly mixed.

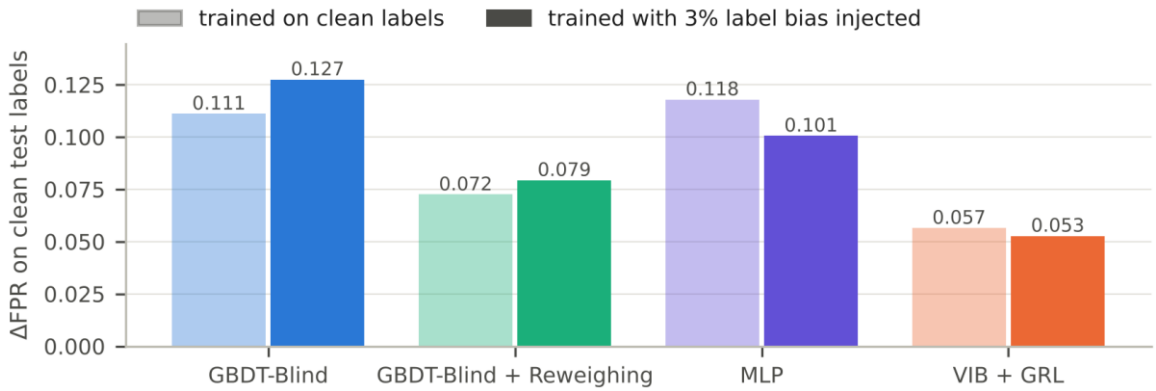


Figure C5: Stress test: ΔFPR on clean test labels after 3% of German and French non-returned training lines were relabelled as returns (one seed).

Appendix D. Reproducibility

Artifacts and Open-Source Code. The complete source code, deterministic verification scripts (eval_deterministic.py), raw DeepSeek API audit traces (g2_audit_raw_trace.jsonl), and evaluation datasets are permanently archived and openly available at: https://github.com/adantn/FairReturns-CDS521-Group14_Doubao Software. Python 3.13 with NumPy 2.5, pandas 3.0, scikit-learn 1.9, SciPy 1.18, Matplotlib 3.11 and Flask 3.1; Graphviz for Figures 2 and 3. The data set [10] is used under its CC BY 4.0 licence. Random seeds are fixed, and all LLM calls are cached in outputs/genai/.

Main results (about 25 minutes on a dual-core laptop):

```
python run_pipeline.py
python run_pipeline.py --sweep --stress
python -m fairreturn.experiments.report_figures
python -m fairreturn.backend
```

Optional live-backend extensions. Set `LLM_BASE_URL`, `LLM_API_KEY` and `LLM_MODEL` of any OpenAI-compatible service in `.env` (the key is never committed), and optionally have two annotators fill `data/processed/human_tag_sample.csv`. All calls are cached in outputs/genai/. Then run:

```
python -m fairreturn.genai.product_tagger --backend api --top 1000
python -m fairreturn.experiments.run_all --product-tags llm \
      --methods gbdt_blind gbdt_notags vib_grl --seeds 3 --bootstrap 0 --quiet
python -m fairreturn.genai.llm_audit --backend api
python -m fairreturn.genai.assistant_eval --backend api --n 100
python -m fairreturn.experiments.report_numbers
```

Results from the offline mock backend are refused by report_numbers. The G2 results of Table 7 come from the official-API run described in Section 5.6.